Lunar Writer / parallel EPROM reader–programmer

USB / CH340C / ATmega / clock / reset / ISP / UART

Open sheet 2

File: controller.kicad_sch

Address generator and NM27C512 direct socket

Open sheet 3

File: address_target.kicad_sch

Data protection and status LEDs

Open sheet 4

File: data_status.kicad_sch

TTL–compatible bidirectional data bus

Open sheet 5

File: ttl_bus.kicad_sch

External VPP / high–side switch / 6.25 V regulator

Open sheet 6

File: programming_power.kicad_sch

Fast VPP pulse switch / indicator / ADC monitor

Open sheet 7

File: programming_pulse.kicad_sch

Default–read relay / target connector / monitoring

Open sheet 8

File: target_interface.kicad_sch

Test points and safety sequence

Open sheet 9

File: testpoints_safety.kicad_sch

2716 / KR573RF5 adapter reference

Open sheet 10

File: adapter_2716_KR573RF5.kicad_sch

28–pin EPROM adapter reference

Open sheet 11

File: adapter_28pin_27xx.kicad_sch

32–pin JEDEC adapter reference

Open sheet 12

File: adapter_32pin_JEDEC.kicad_sch

12–24V protection / boost / regulated VPP

File: writer_supply.kicad_sch

LUNAR WRITER / USB 5V LOGIC / SEPARATE 12–24V INPUT / RESET = READ

Boost generates ~30V from 12–24V input; VPP rails are adjusted separately and selected by JP1.
Direct NM27C512 socket uses 12.75V VPP / 6.25V VCC. Other EPROMs require verified profiles and adapters.
Default firmware locks WRITE until measured commissioning. One target at a time.

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

Sheet: /
File: Lunar_Writer.kicad_sch

Title: Lunar Writer – Rev. 1

Size: A2	Date: 2026–10–12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/13

USB / CH340C / ATmega / clock / reset / ISP / UART

USB input and internal-clock USB UART

J1
USB Micro-B

F1
500mA hold

PWR_FLAG

U5
CH340C

R4
1k

R5
1k

C1
10uF / 10V

C2
100nF

C3
100nF

CH340C: no 12 MHz crystal. R232 = GND (TTL mode).
Pins 7/8 are variant-dependent: leave unconnected.
Only DTR# (13) is used; other modem pins are NC.

J2
AVR ISP

J2: 1 MISO, 2 +5V, 3 SCK,
4 MOSI, 5 RESET, 6 GND.
Never power J2 and USB from competing sources.

J3
UART (MCU RX/TX)

J3 names are from MCU viewpoint.
1 = GND 2 = RX (PD0) 3 = TX (PD1)
Use 5 V TTL UART; avoid simultaneous UART drivers.

ATmega328P-AU / 16 MHz / 5 V

U1
ATmega328P-AU

Y1
16MHz, CL -18pF

R1
10k

SW1
RESET

C11
22pF

C12
22pF

C13
100nF auto-reset

D9
BAT54WS RESET clamp

C4
100nF

C5
100nF

C6
100nF

C10
100nF

D9 clamps the positive DTR edge on RESET, protecting the 5V interlock inputs.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

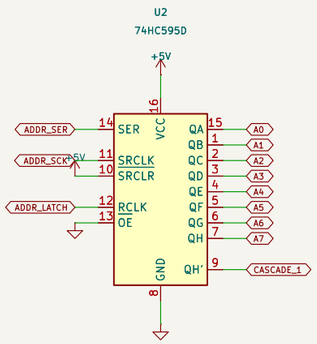
Sheet: /USB / CH340C / ATmega / clock / reset / ISP / UART/
File: controller.kicad_sch

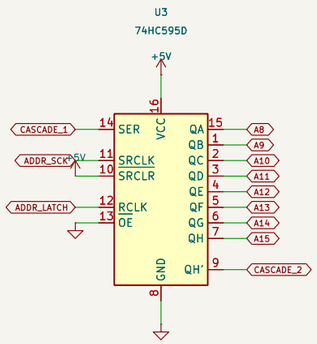
Title: Lunar Writer – Rev. 1

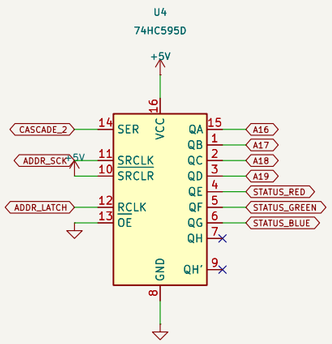
Size: A3 | Date: 2026-10-12 | **Rev: 1**
KiCad E.D.A. 10.0.6 | Id: 2/13

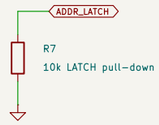
Address generator and NM27C512 direct socket

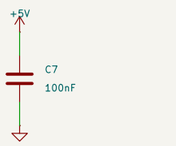
Address / status shift registers: SPI shifts while latch stays LOW

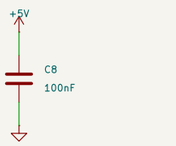


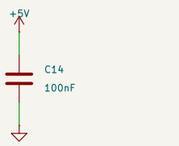




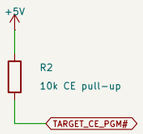


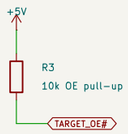


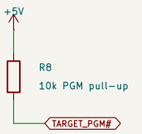


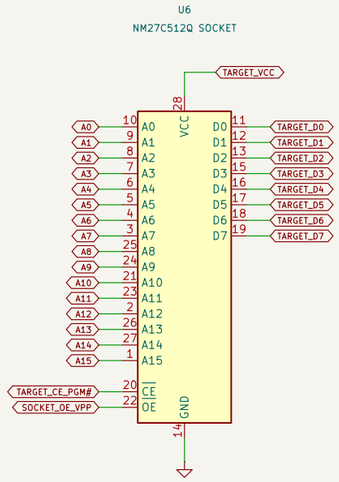


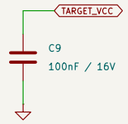
Send U4 byte first, U3 byte second, U2 byte last; MSB first.
U2 Q0–Q7 = A0–A7; U3 Q0–Q7 = A8–A15.
Stage 2: U4 Q0–Q3 = A16–A19; Q4/Q5/Q6 = active-low R/G/B.
OE# = GND and SRCLR# = +5V on every 595.

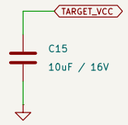












Direct socket U6 is dedicated to NM27C512.
One target only: socket OR adapter.
Pin 20 = CE/PGM#; pin 22 = OE/VPP via K1.
READ: +5V, OE logic. PROG: +6.25V, switched VPP.
Do not insert/remove a target with USB or VPP connected.

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

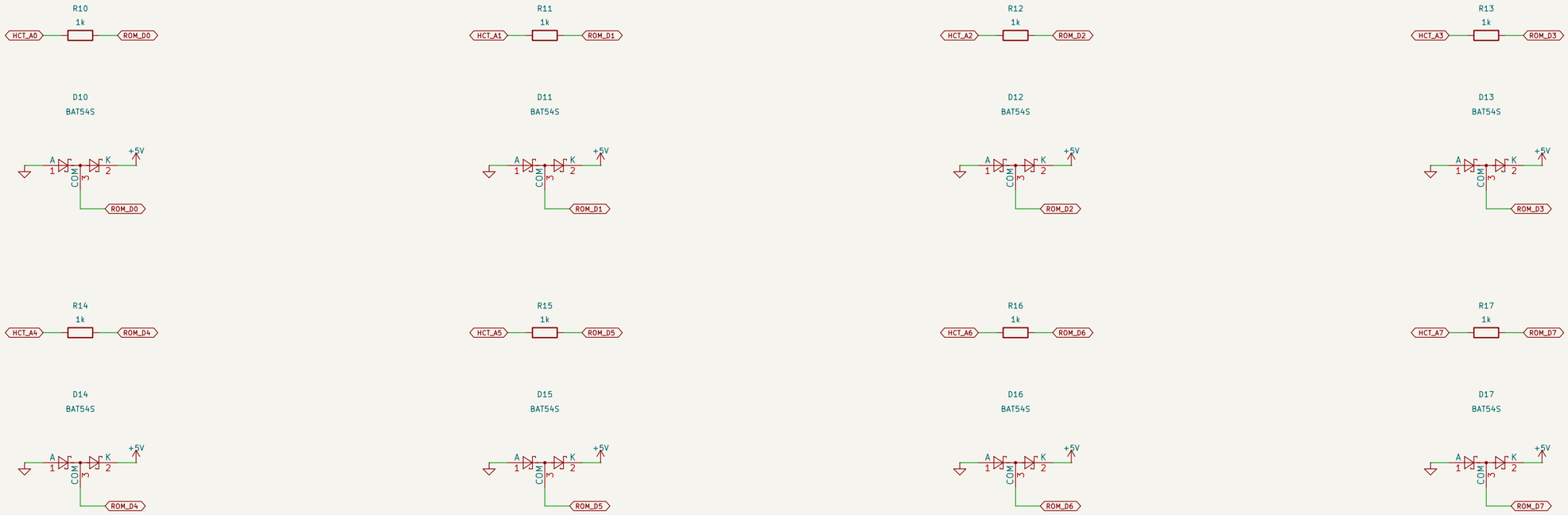
Sheet: /Address generator and NM27C512 direct socket/
File: address_target.kicad_sch

Title: Lunar Writer – Rev. 1

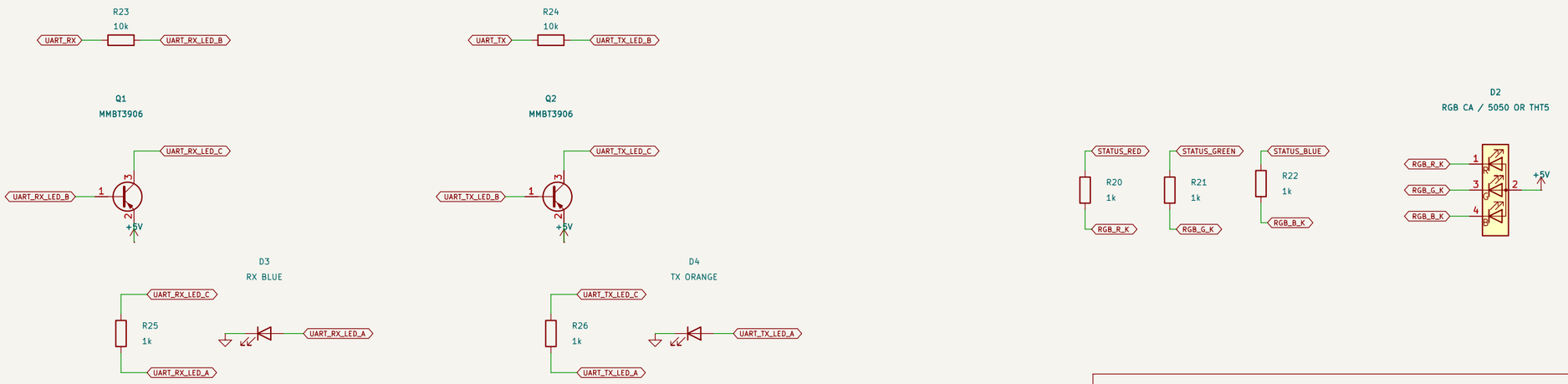
Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 3/13

Data protection and status LEDs

Eight bidirectional data paths: series current limit + external Schottky clamps



BAT54S: pin 1 anode = GND; pin 2 cathode = +5V; pin 3 midpoint = MCU data.
High data clamps toward +5V; negative data clamps toward GND. No target HV on these paths.
1k limits MCU/buffer contention. HCT245 and target-side protection are on the TTL bus sheet.



RGB cathode LOW = ON. Green blink 500ms: READ; Purple blink 500ms: WRITE; Amber: READY; Red: disconnected.
RX/TX PNP emitters = +5V; base current is limited by 10k. UART HIGH idle = LEDs OFF.
D2: combined SMD 5050 / THT5 footprint. Populate ONE ONLY; pads 1R/2A/3G/4B.

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

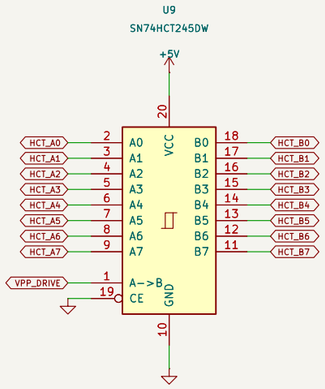
Sheet: /Data protection and status LEDs/
File: data_status.kicad_sch

Title: Lunar Writer — Rev. 1

Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 4/13

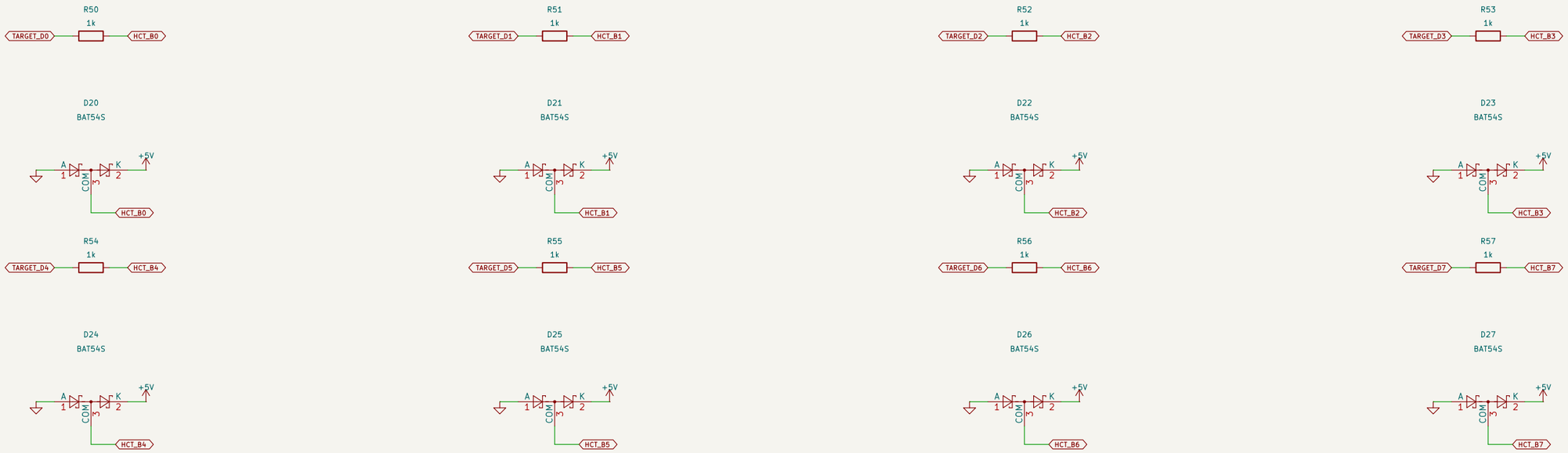
TTL-compatible bidirectional data bus

TTL-compatible bidirectional buffer: preserves reads from older 27xx EPROMs





DIR = VPP_DRIVE: LOW -> target-to-MCU (READ/VERIFY); HIGH -> MCU-to-target (PROGRAM).
OE# tied LOW. Firmware releases MCU data outputs BEFORE removing VPP.
74HCT input VIH = 2.0V; MCU sees full 5V logic. Do not substitute 74HC245.
MCU-side R10-R17 and D10-D17 are on data/status sheet. Target-side protection below.



Target-side clamps keep +6.25V verify outputs away from the 5V buffer. Maximum nominal injection ~1mA/line.
USB-powered logic and relay draw more current than all eight clamps inject; never operate with USB absent.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

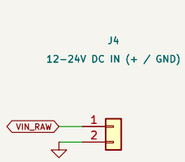
Sheet: /TTL-compatible bidirectional data bus/
File: ttl_bus.kicad_sch

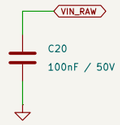
Title: Lunar Writer — Rev. 1

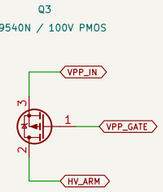
Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 5/13

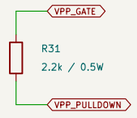
External VPP / high-side switch / 6.25 V regulator

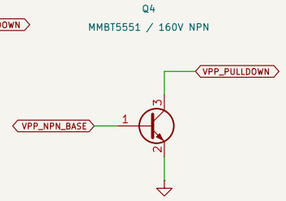
J4: 12...24 V DC input. Boost / VPP regulation on the supply sheet. USB powers logic only.

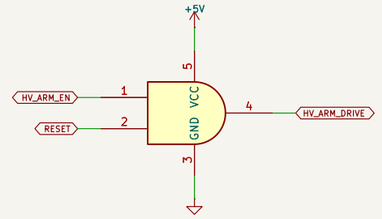




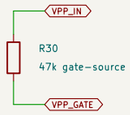


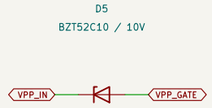




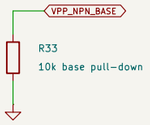


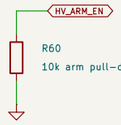






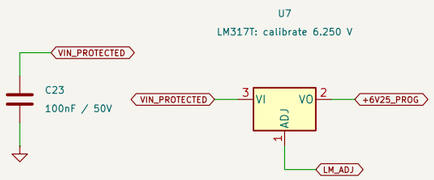


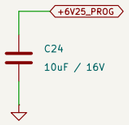


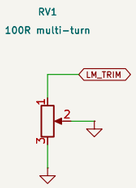


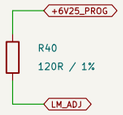


Gate Zener: cathode to SOURCE/regulated VPP_IN, anode to GATE.
Q4 collector pulls gate through R31; emitter = GND.
RESET LOW forces U10 output LOW immediately.
PB1 is the hardware arming interlock; PD2 controls the final pulse switch.

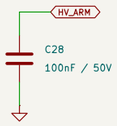


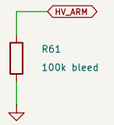


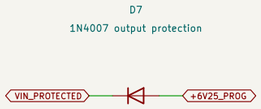












Set TP_6V25 to 6.250V at load. U7 runs from VIN_PROTECTED.
U7 tab is OUTPUT; fit an insulated heatsink. Worst nominal loss at 24V/100mA: ~1.8W.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

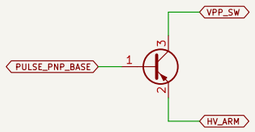
Sheet: /External VPP / high-side switch / 6.25 V regulator/
File: programming_power.kicad_sch

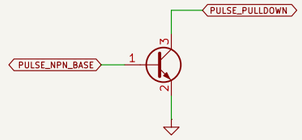
Title: Lunar Writer - Rev. 1

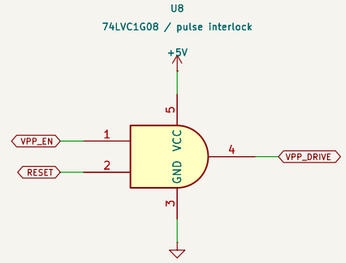
Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 6/13

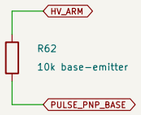
Fast VPP pulse switch / indicator / ADC monitor

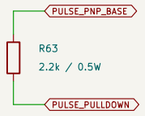
Final VPP pulse switch: separate slow arming from pin22 rise/verify switching



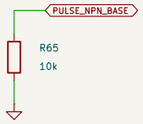


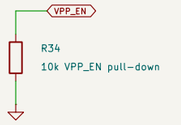




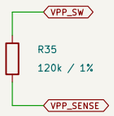


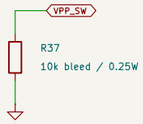


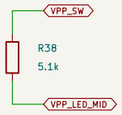


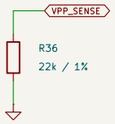


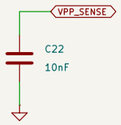


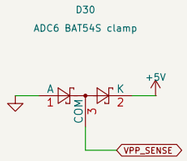


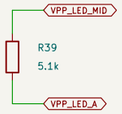


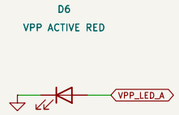












Arm Q3 while relay remains READ; settle HV_ARM; then pulse Q7 using PD2.
Preflight measures actual switched VPP with socket isolated. Remove VPP before moving relay.
PNP emitter = HV_ARM, collector = VPP_SW. VPP has a load-dependent VCE drop: calibrate under 30mA load.
No added VPP_SW capacitor. R37 returns pin22 LOW for byte verify. ADC filter is behind 120k.
Commissioning requires measured <=50ns pin22 rise time and <14V overshoot for NM27C512.
WRITE is locked in the default firmware build until the assembled pulse stage has passed these checks.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

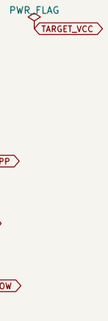
Sheet: /Fast VPP pulse switch / indicator / ADC monitor/
File: programming_pulse.kicad_sch

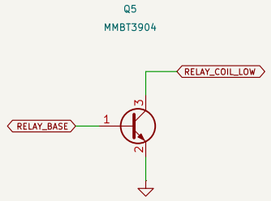
Title: Lunar Writer - Rev. 1

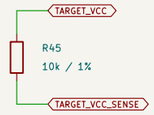
Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 7/13

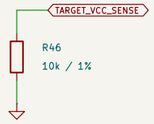
Default—read relay / target connector / monitoring

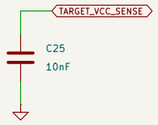
K1 DPDT / break—before—make / default READ position

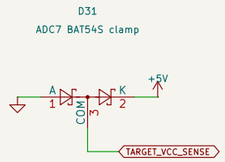


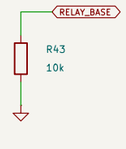






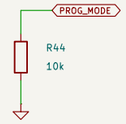








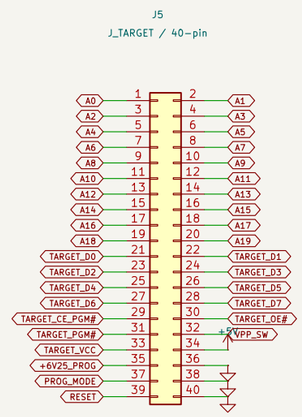
ADC7: half of target VCC.
5.00V —> 2.50V; 6.25V —> 3.125V.





Coil 1/16. Pole A: COM4, NC6, NO8. Pole B: COM13, NC11, NO9.
De—energized: target VCC = +5V, socket OE/VPP = MCU OE#.
Energized: target VCC = +6.25V, socket OE/VPP = VPP_SW.
470R base resistor supplies ~9mA: the standard G5V—2 coil draws 100mA.
The suggested 10k base drive would not guarantee saturation.

J_TARGET: odd pins on left; even pins on right.
1—20 A0—A19; 21—28 D0—D7 (protected target side);
29 CE/PGM#, 30 OE#, 31 PGM#, 32 VPP_SW;
33 TARGET_VCC, 34 +5V, 35 +6.25V, 36 GND;
37 PROG_MODE, 38 GND, 39 RESET, 40 GND.
NEVER wire VPP_SW to address/data/control GPIO.
Adapters requiring VPP=VCC in READ need physical switching.
Keep onboard U6 empty when an adapter is installed.



USB 5V logic / separate 12—24V DC input / regulated VPP / default READ

Sheet: /Default—read relay / target connector / monitoring/
File: target_interface.kicad_sch

Title: Lunar Writer — Rev. 1

Size: A3	Date: 2026—10—12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 8/13

Test points and safety sequence

TP1
TP_5V

TP2
TP_GND

TP3
TP_6V25



TP4
TP_VPP_IN



TP5
TP_VPP_SW



TP6
TP_RESET



TP7
TP_ADDR_SCK



TP8
TP_ADDR_LATCH



TP9
TP_CE



TP10
TP_OE



TP11
TP_PGM



TP12
TP_RX



TP13
TP_TX



Safety sequence / required bench checks

1. USB supplies all 5V logic. External supply is regulated, current-limited and shares GND.
2. Power-up / RESET / ISP / DTR reset: CE and OE HIGH, relay READ, VPP disabled.
3. Preflight VPP measurement is made with relay in READ; onboard socket is isolated from VPP.
4. Verify target VCC before enabling VPP. Verify VPP voltage before every programming pulse.
5. Remove VPP and confirm VPP_SW <0.40V before verifying or releasing PROG_MODE.
6. Watchdog / timeout / invalid command returns to safe mode.
7. NEVER insert/remove an EPROM with USB or VPP connected. One populated target only.
8. Before commissioning WRITE, scope pin 22 for overshoot (<14V) and rise time (<=50ns).
9. This revision is a schematic/prototype basis; ERC cannot certify switching waveforms or PCB layout.

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

Sheet: /Test points and safety sequence/
File: testpoints_safety.kicad_sch

Title: Lunar Writer – Rev. 1

Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 9/13

2716 / KR573RF5 adapter reference

REFERENCE ONLY — logical mapping for future adapter PCBs, not additional live targets

Pin 01: A7

Pin 02: A6

Pin 03: A5

Pin 04: A4

Pin 05: A3

Pin 06: A2

Pin 07: A1

Pin 08: A0

Pin 09: D0

Pin 10: D1

Pin 11: D2

Pin 12: GND

Pin 13: D3

Pin 14: D4

Pin 15: D5

Pin 16: D6

Pin 17: D7

Pin 18: CE/PGM (READ low)

Pin 19: A10

Pin 20: OE#

Pin 21: VPP (READ +5V)

Pin 22: A9

Pin 23: A8

Pin 24: TARGET_VCC

Intel 2716: VPP pin21 must be +5V for READ. NOT VPP_SW=0V.
Read adapter ties pin21 to TARGET_VCC; pin18 = CE, pin20 = OE.
Future WRITE needs a relay/jumper to select pin21 between +5V and VPP_SW;
separate exact algorithm and polarity are required. Do not copy NM27C512 WRITE.
KR573RF5 profile is quarantined pending a traceable original device datasheet.
Its manufacturer pinout/algorithm must be checked before populating an adapter.
2732 differs: pin21 = A11 and pin20 = OE/VPP (requires HV isolation).

Source: Intel 2716 datasheet, Intel 1980, pin configuration / mode selection.
See docs/sources.md for the source URL. No unverified KR573RF5 wiring is fabricated.

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

Sheet: /2716 / KR573RF5 adapter reference/
File: adapter_2716_KR573RF5.kicad_sch

Title: Lunar Writer — Rev. 1

Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 10/13

28-pin EPROM adapter reference

REFERENCE ONLY - logical mapping for future adapter PCBs, not additional live targets

Pin 01: 64/128/256: VPP; 512: A15

Pin 02: A12

Pin 03: A7

Pin 04: A6

Pin 05: A5

Pin 06: A4

Pin 07: A3

Pin 08: A2

Pin 09: A1

Pin 10: A0

Pin 11: D0

Pin 12: D1

Pin 13: D2

Pin 14: GND

Pin 15: D3

Pin 16: D4

Pin 17: D5

Pin 18: D6

Pin 19: D7

Pin 20: 64/128: CE#; 256/512: CE/PGM#

Pin 21: A10

Pin 22: 64/128/256: OE#; 512: OE/VPP

Pin 23: A11

Pin 24: A9

Pin 25: A8

Pin 26: 64: NC; 128/256/512: A13

Pin 27: 64/128: PGM#; 256/512: A14

Pin 28: TARGET_VCC

2764/27128: pin27 is PGM#. NOT A14. 27256: pin27 is A14.
Do not route a 595 output onto a VPP pin. Use keyed jumpers or a device-specific adapter.
VPP on pin1 needs READ +5V selection, separate from the onboard NM27C512 relay.
Read-only adapters may tie VPP and unused PGM# HIGH. WRITE remains disabled for these profiles.
Sources: National NM27C512 and Intel 2764 / 27128A / 27256 original datasheets.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

Sheet: /28-pin EPROM adapter reference/
File: adapter_28pin_27xx.kicad_sch

Title: Lunar Writer - Rev. 1

Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 11/13

32-pin JEDEC adapter reference

REFERENCE ONLY — logical mapping for future adapter PCBs, not additional live targets

Pin 01: 010/020/040: VPP; 080: A19

Pin 02: A16

Pin 03: A15

Pin 04: A12

Pin 05: A7

Pin 06: A6

Pin 07: A5

Pin 08: A4

Pin 09: A3

Pin 10: A2

Pin 11: A1

Pin 12: A0

Pin 13: D0

Pin 14: D1

Pin 15: D2

Pin 16: GND

Pin 17: D3

Pin 18: D4

Pin 19: D5

Pin 20: D6

Pin 21: D7

Pin 22: CE / CE-PGM# (family dependent)

Pin 23: A10

Pin 24: 010/020/040: OE#; 080: OE/VPP

Pin 25: A11

Pin 26: A9

Pin 27: A8

Pin 28: A13

Pin 29: A14

Pin 30: 010: NC; 020/040/080: A17

Pin 31: 010/020: PGM#; 040/080: A18

Pin 32: TARGET_VCC

National 27C010/020/040/080 package comparison is the source for these pin roles.
Use a device-specific adapter; role changes require keyed selection or HV-rated switching.
Pin1 must never receive VPP when it is A19. Pin24 changes from OE to OE/VPP on 080.
27C080 cannot use a generic 32-pin VPP-pin1 adapter. Program profiles remain disabled.
Read VPP pins require +5V, not an unpowered VPP_SW. See device_profiles.json for verified status.

USB 5V logic / separate 12-24V DC input / regulated VPP / default READ

Sheet: /32-pin JEDEC adapter reference/
File: adapter_32pin_JEDEC.kicad_sch

Title: Lunar Writer — Rev. 1

Size: A3	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 12/13

Lunar Writer: protected 12–24V supply, boost and VPP rails

External power is never connected to USB VBUS / +5V. J4 is on the power sheet.

F2
MF-R090 / 0.9A hold / 60V
VIN_RAW
VIN_FUSED

D40
SS36 / 3A / 60V reverse-polarity
VIN_FUSED
VIN_PROTECTED

D41
SMBJ24A / TVS
VIN_PROTECTED

L1
100uH / SRR1210A-101M / Isat 2.2A
VIN_PROTECTED
BOOST_SW

U11
LM2577T-ADJ / 40V input, 60V switch
VIN_PROTECTED
BOOST_COMP
BOOST_FB
VIN
SW
COMP
FB
GND
BOOST_SW

D42
SS36 / 3A / 60V boost rectifier
BOOST_SW
BOOST_30V

C32
680uF / 50V low ESR
BOOST_30V

R70
47k / 1%
BOOST_30V
BOOST_FB

R71
2k / 1%
BOOST_FB

C33
100nF / 50V X7R
BOOST_30V

TP15
BOOST_30V
BOOST_30V

C30
220uF / 50V low ESR
VIN_PROTECTED

C31
100nF / 50V X7R
VIN_PROTECTED

TP14
VIN_PROTECTED
VIN_PROTECTED

R72
470R compensation
BOOST_COMP
BOOST_COMP_RC

C34
47uF / 16V compensation
BOOST_COMP_RC

Vboost = 1.23*(1+47k/2k) = 30.135V. L=100uH; Rc=470R; Cc=47uF; Cout=680uF.
Design load 100mA on 30V rail. Use 50V bulk capacitors. Cout ESR <=0.2ohm / ripple >=0.2A.
Validate startup and load-step response; calculated compensation is not a measured stability test.

U12
LM317T / 12.75V / heatsink
BOOST_30V
VI
VO
ADJ
VPP_12V75
VPP_ADJ12

R73
120R / 1%
VPP_12V75
VPP_ADJ12

R74
1k / 1%
VPP_ADJ12
VPP_TRIM12

RV2
200R multi-turn
VPP_TRIM12

D43
M7 / 1N4007-SMD / 1A 1000V
VPP_12V75
BOOST_30V

C35
1uF / 50V X7R
VPP_12V75

TP16
VPP_12V75
VPP_12V75

U13
LM317T / 21...25V / heatsink
BOOST_30V
VI
VO
ADJ
VPP_HIGH
VPP_ADJ13

R75
120R / 1%
VPP_HIGH
VPP_ADJ13

R76
1.8k / 1%
VPP_ADJ13
VPP_TRIM13

RV3
1k multi-turn
VPP_TRIM13

D44
M7 / 1N4007-SMD / 1A 1000V
VPP_HIGH
BOOST_30V

C36
1uF / 50V X7R
VPP_HIGH

TP17
VPP_HIGH
VPP_HIGH

JP1
VPP SELECT: 12.75 / common / HIGH
VPP_12V75
VPP_IN
VPP_HIGH

R77
100k bleed
VPP_IN

JP1: fit ONE shunt only: 1-2 = NM27C512 12.75V, 2-3 = adjusted HIGH for a verified adapter.
No shunt = VPP disconnected / discharged. Do not change JP1 with either supply powered.
RV2: adjust actual VPP_5W to 12.75V under load. RV3: adjust HIGH to the exact device voltage.
U12/U13 tabs are OUTPUTS: separate insulated heatsinks. Programming still: defaults OFF on RESET.

PWR_FLAG
VIN_PROTECTED

PWR_FLAG
BOOST_30V

PWR_FLAG
VPP_IN

USB 5V logic / separate 12–24V DC input / regulated VPP / default READ

Sheet: /12–24V protection / boost / regulated VPP/
File: writer_supply.kicad_sch

Title: Lunar Writer – Rev. 1

Size: A2	Date: 2026-10-12	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 13/13